

PrivAnalogy: An Analogy Mechanism-Based Privacy Protection Framework for LLM Prompts

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Abstract—During interactions with large language models (LLMs), user-submitted prompts may contain sensitive information, thereby introducing potential privacy risks. Given the inherent complexity of LLMs, minor changes in input can significantly influence output behavior. This presents a critical challenge in achieving effective privacy protection while maintaining model utility. To address this problem, this study introduces *PrivAnalogy*, a privacy-preserving framework that utilizes an *Analogy mechanism* on the client side to semantically transform sensitive data within prompts, thereby preventing cloud-based LLMs from directly accessing such content. The framework comprises four functional modules, with *Analogy selection* and *Analogy reversion* serving as the core components. These two modules are primarily responsible for the semantic transformation of sensitive information and the contextual restoration of the original intent. The analogy selection module applies local differential privacy principles to transform sensitive content into analogous expressions, ensuring that the LLM service only processes privacy-preserving input. Subsequently, the analogy reversion module performs semantic alignment between the generated response and the original prompt, ensuring that the final response reliably reflects the user's intended query. Comprehensive framework evaluations across three datasets show that *PrivAnalogy* achieves an average resistance to Prompt inversion attacks that is $1.79\times$ stronger than InferDPT + SANTEXT+, $1.37\times$ stronger than InferDPT + RANTEXT, and $1.65\times$ stronger than InferDPT + CUSTEXT+, while consistently maintaining reliable response quality for end users.

Index Terms—Differential privacy, LLM privacy protection, analogy mechanism, prompt-based interaction.

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The source code of *PrivAnalogy* is publicly available at <https://github.com/superMrsong/PrivAnalogy>.

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I. INTRODUCTION

IN RECENT years, LLM services have been widely adopted across various domains, such as personal productivity, everyday support, and office assistance, significantly enhancing user experience [1]. However, this development of LLM services has also raised growing concerns regarding the risk of privacy breaches [2], [3]. Given the advanced language understanding and information integration capabilities of LLMs, users are often required to provide detailed contextual descriptions within prompts to obtain accurate and reliable responses [4]. These inputs frequently contain not only sensitive personally identifiable information (PII) [5], but also implicit behavioral patterns, preferences, and contextual details, which LLMs may further infer through advanced semantic reasoning [6]. For instance, a prompt such as “Patient John, a 35-year-old male, is a car cleaning worker who was infected with schistosomiasis and developed dizziness, tinnitus, and other symptoms. How should he be treated?” illustrates how users often need to disclose specific personal and contextual information to obtain useful recommendations [7]. During interactions with LLM services, prompts are received, processed, and stored by service providers, thereby posing risks of sensitive information leakage beyond the user's control [8].

As illustrated in Fig. 1, this study considers a scenario in which users interact with a cloud-based LLM service via textual prompts. These prompts may contain sensitive personal information, such as lifestyle patterns, occupational details, and private attributes. Once submitted to the cloud, such information becomes exposed to various privacy threats, including prompt injection during submission, membership inference during inference, data poisoning through retained logs, and model extraction and inversion through repeated queries. These threats indicate that prompt privacy cannot be fully protected once sensitive content leaves the local environment without additional safeguards.

Although many privacy-preserving techniques have been studied, protecting prompt-level privacy without degrading chat quality remains challenging. Existing approaches commonly rely on privacy-preserving data publishing and NLP techniques, including homomorphic encryption (HE) [13], data anonymization [14], and differential privacy (DP) [15]. HE provides strong privacy guarantees but is too computationally expensive for real-time LLM interaction [16]. Data anonymization is efficient, but it often fails in black-box LLM settings because strong contextual modeling can help reconstruct sensitive user information [8]. DP offers a rigorous way to limit the memorization and disclosure

TABLE I

COMPARISON OF METHODS FOR PROTECTING USER PRIVACY IN LLM INTERACTIONS. ATTACK ABBREVIATIONS: PIA = PROMPT INJECTION ATTACKS; MIA = MEMBERSHIP INFERENCE ATTACKS; DPL = DATA POISONING VIA LOGS; MEI = MODEL EXTRACTION & INVERSION ATTACKS. A CHECKMARK (✓) INDICATES THAT THE METHOD'S THREAT MODEL ADDRESSES THE CORRESPONDING ATTACK (SEE FIG. 1)

Method	Protected Subject	Triggering Phases	Target Attacks			
			PIA	MIA	DPL	MEI
DP-OPT [9]	PII	Before prompt submission	✓			
DePrompt [7]	Named entities	Before prompt submission	✓			
Rap-LI [10]	All tokens	Before prompt submission	✓	✓		
HaS [11]	Named entities	Before prompt submission and after LLM response	✓	✓		
InferDPT [12]	All tokens	Before prompt submission and after LLM response	✓	✓		
<i>PrivAnalogy</i> (Our)	Semantic association	Before prompt submission and after LLM response	✓	✓	✓	✓

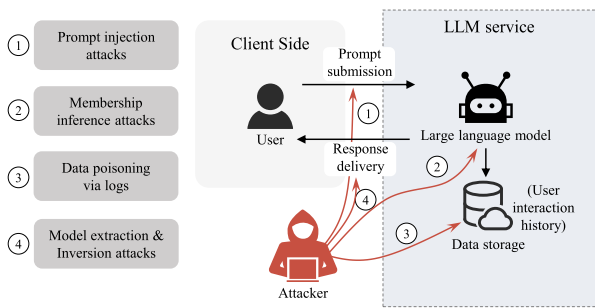


Fig. 1. Overview of the threat model illustrating potential privacy risks during user-LLM interactions, including adversarial prompting, data poisoning, and inference attacks.

of personal data, making it a promising strategy for privacy protection in LLM services under practical conditions [17].

Two major challenges arise when applying DP to black-box LLM services. First, most existing DP-based prompt protection methods, such as SANTEXT+ [18], CUSTEXT+ [19], and RANTEXT [12], rely on token-level perturbation to achieve privacy guarantees. These methods modify prompts before they are submitted to LLMs. However, LLMs are highly sensitive to input changes [20]. Token substitutions may alter the original meaning of a prompt. They may also disrupt grammatical structure and logical consistency. As a result, sanitized prompts can become difficult for LLMs to interpret correctly. This often reduces response quality and conversational coherence [21], [22], [23].

Second, existing DP-based approaches primarily protect data by perturbing local tokens. Such mechanisms do not fully capture privacy risks arising from broader contextual information. Recent studies show that sensitive attributes can still be inferred from model outputs despite DP-based sanitization [24]. This problem becomes more pronounced in black-box settings, where attackers only observe generated responses. These challenges highlight the need for privacy-preserving mechanisms that maintain prompt usability while providing stronger protection against inference attacks.

To address the challenges above, this paper proposes *PrivAnalogy*, a privacy protection framework tailored for black-box LLM interaction scenarios. As summarized in Table I, existing

representative LLM privacy techniques are compared based on their key features. *PrivAnalogy* is the first to identify semantic associations as protected subjects and transform them through analogy selection while preserving logical coherence. The overall prompt protection procedure, including sensitive association identification and analogy selection, is referred to as *PromptAnalogy*. Upon receiving responses from the LLM service, *PrivAnalogy* applies analogy reversion and response refinement models to align the response—which integrates knowledge from the LLM—with the user’s original prompt, ultimately generating replies that meet user expectations. Since sensitive data is processed entirely locally, the framework effectively prevents external exposure and mitigates privacy leakage at the source and during response delivery while ensuring reliable service for users.

Our key contributions are as follows:

- *First proposal of the Analogy mechanism*: This study introduces an innovative text semantic prediction-based analogy mechanism that performs analogy transformation of sensitive information through deep abstraction and semantic transfer. Furthermore, it aligns the LLM’s response with the user’s prompt to ensure semantic consistency and reliability.
- *Proposed PrivAnalogy framework*: This study is the first to define semantic association as protected subjects and introduces a privacy protection framework covering the entire process from prompt construction to response generation. Built upon the analogy mechanism and a local LLM, the framework effectively safeguards user privacy while maintaining the logical integrity of prompts. It also ensures the usability and logical consistency of final responses, achieving a balance between privacy protection and practical utility.

II. RELATED WORK

A. Privacy Preservation for User Prompts in LLM Services

Recent studies have highlighted the privacy risks associated with interactions with LLM services. [8] show that modern LLMs can infer sensitive user attributes, such as location, income, and gender, from seemingly innocuous text. [25] further

demonstrate that users frequently disclose PII [26], [27] when interacting with cloud-based LLM services. These findings have motivated extensive research on prompt privacy protection mechanisms.

1) *Non-DP Approaches*: Most non-DP approaches protect prompt privacy through heuristic or LLM-driven techniques. HaS [11], PP-TS [28], and Casper [29] represent the dominant entity-level sanitization paradigm. These methods identify sensitive entities and replace them before prompts are sent to external LLMs. They typically rely on rule-based techniques, NER models, or lightweight local classifiers. Portcullis [30] extends this idea through a privacy gateway that sanitizes prompts before model access. Pr ϵ mpt [31] adopts a cryptography-inspired prompt transformation mechanism to obfuscate sensitive content. More recently, [32] use LLMs themselves as anonymizers. Their framework iteratively removes privacy-relevant information through adversarial interactions between LLMs while preserving task utility.

Despite their practical value, these methods remain limited in several ways. Because most approaches rely on heuristic protection strategies, their effectiveness largely depends on the accurate detection of sensitive information. When sensitive attributes are implicit or distributed across contextual cues, these transformations may leave privacy-relevant evidence in the processed prompts. This risk is further exacerbated by the absence of formal privacy guarantees, since adversaries with auxiliary knowledge may still infer sensitive information from sanitized prompts or model outputs [8], [26].

2) *DP-Based Approaches*: DP-based approaches provide formal privacy guarantees under well-defined threat models. Text sanitization methods reduce privacy leakage by modifying sensitive information through removal, replacement, or perturbation mechanisms. SANTEXT+ [18] replaces sensitive tokens with semantically similar alternatives selected by a DP mechanism, while CUSTEXT+ [19] introduces a non-sensitive token dictionary to guide the sanitization process. TextObfuscator [33] protects privacy through prototype learning and embedding perturbation, whereas RANTEXT [12] constructs candidate lists based on perturbation radii to improve privacy protection in practical scenarios. Overall, these methods still struggle to balance privacy and utility. Replacing tokens with semantically similar alternatives in the embedding space provides limited resistance to inference attacks, while stronger perturbation improves privacy but may reduce the utility of the final output.

Recent studies have advanced DP-based sanitization and explored its application to privacy-preserving LLM inference. CluSanT [34] improves DP-based text sanitization by restricting token replacements within semantically related clusters. InferDPT [12] integrates DP sanitization methods, including RANTEXT, SANTEXT+, and CUSTEXT+, into black-box LLM inference. Rap-LI [10] introduces risk-aware privacy protection by adaptively assigning privacy levels to different prompt components. However, these approaches mainly focus on protecting sanitized inputs through token-level or component-level modifications, while the impact of input sanitization on the semantic consistency of LLM-generated responses is not explicitly considered.

PrivAnalogy adopts differential privacy to provide formal privacy guarantees. Unlike existing DP-based sanitization methods that rely on token-level removal, replacement, or perturbation, *PrivAnalogy* introduces an *Analogy mechanism* to transform associations within prompts for privacy protection. Moreover, unlike existing LLM privacy-preserving frameworks that mainly focus on input protection, *PrivAnalogy* considers prompt-response coupling by protecting sensitive prompts while maintaining semantic alignment with responses.

III. PRELIMINARIES

Definition 1 (ϵ -Local differential privacy [35]): ϵ -Local Differential Privacy (ϵ -LDP) is a variant of differential privacy in which the server is considered untrusted, and all privacy-preserving computations are executed locally on the client side. Given a privacy parameter $\epsilon \geq 0$, a randomized mechanism M satisfies ϵ -LDP if, for any pair of inputs $x, x' \in X$ and any possible output $y \in Y$, the following condition holds:

$$\frac{\Pr[M(x) = y]}{\Pr[M(x') = y]} \leq e^\epsilon, \quad (1)$$

where ϵ denotes the privacy budget; a smaller value indicates a stronger privacy guarantee, typically at the expense of reduced data utility.

Definition 2 (Exponential mechanism [36]): Given a scoring function $u : X \times Y \rightarrow \mathbb{R}$, a randomized mechanism $M(\cdot)$ satisfies ϵ -LDP if, for any input $x \in X$ and any possible output $y \in Y$, the selection probability is given by:

$$\Pr[y | x] \propto \exp\left(\frac{\epsilon \cdot u(x, y)}{2\Delta u}\right). \quad (2)$$

The global sensitivity Δu is defined as:

$$\Delta u = \max_{x, x' \in X, y \in Y} |u(x, y) - u(x', y)|. \quad (3)$$

The specific form of the utility function u depends on the application context and can be defined or adjusted according to task-specific requirements. To meet the computational constraints of differential privacy, it is common practice to normalize Δu by bounding u . The privacy budget ϵ governs the trade-off between privacy protection and output utility.

IV. PRIVANALOGY

This study proposes *PrivAnalogy* to protect user privacy in LLM service scenarios. Fig. 2 shows that *PrivAnalogy* contains four components: sensitive association identification, analogy selection, analogy reversion, and response refinement.

The sensitive association identification module extracts latent semantic associations from prompts and treats privacy-relevant associations as targets for protection. Based on these targets, the analogy selection module uses an analogy-based mechanism to transform sensitive associations into protected analogical expressions. Together, these two components form the *PromptAnalogy* method, which provides formal privacy protection through DP while preserving contextual coherence.

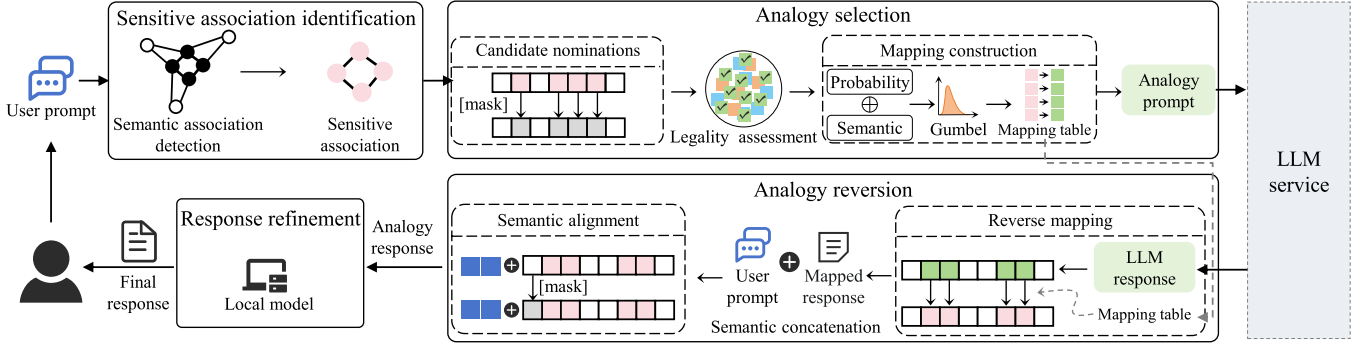


Fig. 2. Overview of the *PrivAnalogy* framework, which consists of four main components: a sensitive association identification module for locating privacy-sensitive positions, an analogy selection module for constructing a DP-protected analogical prompt, an analogy reversion module for restoring the LLM-generated response, and a response refinement module for refining the restored response into the final output.

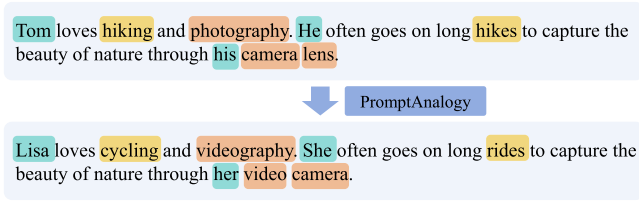


Fig. 3. An illustrative example of prompt transformation using *PromptAnalogy* for privacy preservation.

To ensure reliable model feedback after prompt protection, *PrivAnalogy* applies analogy reversion to map responses generated from protected prompts back to the original semantics, mitigating semantic drift introduced by prompt transformation. It then performs response refinement to improve the restored response and produce the final user-facing output. The implementation details of *PrivAnalogy* are presented next.

A. Sensitive Association Identification

A semantic association refers to a cluster of tokens that are strongly related in meaning and contextually connected. As shown in Fig. 3, in the prompt “Tom loves hiking and photography. He often goes on long hikes to capture the beauty of nature through his camera lens.” the token groups [Tom, he, his], [hiking, hikes], and [photography, camera, lens] can be treated as semantic associations. Once a token within such an association is exposed, an attacker may infer privacy-sensitive attributes, such as user interests or behavioral patterns. Therefore, *PrivAnalogy* treats semantic associations as protection targets for sensitive information.

Given a user prompt X , tokens excluding punctuation, determiners, prepositions, and particles are selected as core tokens to form the set $X' = \{x_1, x_2, \dots, x_n\}$. Based on this set, a semantic co-occurrence graph $G = (V, E)$ is constructed, where the vertex set $V = X'$ and the edge set E represent semantic correlations between tokens. For any two tokens $x_i, x_j \in X'$, if their semantic similarity is high, a weighted edge is added between them in G , with the edge weight defined as:

$$w_{ij} = \cos(\vec{x}_i, \vec{x}_j), \quad (4)$$

where $\vec{x}_i$ denotes the embedding vector of token x_i , and $\cos(\cdot, \cdot)$ represents the cosine similarity function. Based on graph G , the semantic grouping function $f_{\text{Group}}(X)$ aims to find an optimal partition $C = \{c_1, c_2, \dots, c_k\}$ by maximizing the sum of intra-group edge weights:

$$f_{\text{Group}}(X) = \arg \max_{c_1 \cup \dots \cup c_k = X'} \sum_{l=1}^k \sum_{x_i, x_j \in c_l} w_{ij}. \quad (5)$$

B. Analogy Mechanism

PrivAnalogy performs analogy transformations on identified sensitive associations to conceal the protected subjects. It first applies a dynamic masking strategy to generate analogy candidates for these subjects. A masked language model generates candidate substitutions, which are filtered using part-of-speech constraints to ensure semantic validity and grammatical correctness. An LDP mechanism is then applied to sample a substitute from the filtered candidates. The resulting privacy-preserved prompt is submitted to a cloud-based black-box LLM for inference. After receiving the generated response, *PrivAnalogy* restores the original semantics using the constructed analogy mapping and the masked language model. This process aligns the response with the user’s original prompt, preserving semantic consistency and the intended meaning.

1) *Analogy Selection*: To protect implicit sensitive information in prompts while preserving semantic coherence and readability, the analogy mechanism identifies suitable analogs for the protected subjects. The process consists of three stages: analogy candidate generation, candidate filtering, and LDP-based substitute selection.

Candidate nominations: Given the set C of identified protected subjects, the corresponding tokens in the original prompt X are masked to generate candidate substitutions that preserve the same semantic associations. Retaining all candidate substitutions incurs substantial computational and storage overhead. Therefore, the framework adopts the LimitedDomain mechanism [37], which restricts the candidate space to the top- k tokens ranked by prediction probability, thereby reducing both memory consumption and computational cost.

Legality assessment: After generating the candidate substitutions, each candidate is assigned a part-of-speech (POS) tag and compared with the corresponding original token. Only candidates with the same POS as the original token are retained, ensuring grammatical correctness and semantic coherence after substitution.

Mapping construction: We define the candidate selection function f_{select} to sample a substitute from the part-of-speech-filtered candidate set C under the constraint of ε -LDP. Based on the selected substitute, an analogy mapping table π is constructed for subsequent semantic restoration. The selection is guided by a utility function $u(w, x)$ that jointly measures semantic relevance and linguistic fluency. The utility function is defined as follows:

$$u(w, x) = \frac{P_{\text{sem}}(w) \cdot P_{\text{LM}}(w|x)}{\sum_{w' \in D} P_{\text{sem}}(w') \cdot P_{\text{LM}}(w'|x)}, \quad (6)$$

where $P_{\text{LM}}(w|x)$ denotes the probability assigned by the masked language model for generating candidate w in the context x , and $P_{\text{sem}}(w)$ represents the relative frequency of words in the candidate set whose semantic prototype is given by lemma(w). The utility function evaluates the overall suitability of each candidate by normalizing the Bayesian product of semantic and contextual likelihoods.

To achieve randomized candidate selection under ε -LDP, we implement the exponential mechanism using the Gumbel-Max trick. Specifically, an independent Gumbel noise term is added to the utility score of each candidate, and the candidate with the highest perturbed utility is selected. This procedure is equivalent to the exponential mechanism in terms of its sampling distribution while providing an efficient implementation. The selection mechanism is formally defined as:

$$f_{\text{select}}(x) = \arg \max_{w \in D} (u(w, x) + Z_w), \quad (7)$$

where the noise term Z_w is independently sampled from a Gumbel(0, σ) distribution with location parameter 0 and scale parameter σ . The scale parameter is chosen such that the induced sampling distribution is equivalent to that of the exponential mechanism. Accordingly, to guarantee ε -LDP, σ is set to $2\Delta u/\varepsilon$. Since the normalized utility function satisfies $u(w, x) \in [0, 1]$, its global sensitivity is conservatively upper-bounded by $\Delta u = 1$.

Theorem 1: The selection mechanism defined in Equation 7 induces the same distribution as the exponential mechanism with utility function u . Consequently, when the scale parameter is set to $2\Delta u/\varepsilon$, it satisfies ε -LDP.

The proof is provided in Appendix A, available online.

2) **Analogy Reversion:** To ensure that the final response accurately reflects the user's original intent, *PrivAnalogy* first applies reverse mapping to the response generated by the black-box LLM using the constructed analogy mapping table, and then combines the restored response with the original prompt. The analogy mechanism is subsequently used to identify and select the tokens that best fit the contextual semantics, thereby restoring semantic consistency between the prompt and the response. This process mitigates semantic drift introduced by prompt

transformation and improves the accuracy and reliability of the final response.

Algorithm 1: Analogy Reversion Algorithm.

Require: Analogy prompt $\tilde{X}$, mapping table π , initial response R generated by LLM

Ensure: Analogy response A

Step 1: Reverse Mapping

1: **for** each mapping pair $(\tilde{w}, w) \in \pi$ **do**
 2: $R' \leftarrow \text{ReplaceAll}(R, \tilde{w}, w)$ // Restore original entities
 3: **end for**

Step 2: Semantic Concatenation

4: $T \leftarrow \tilde{X} \oplus R'$ // Concatenate prompt and response

Step 3: Semantic Alignment

5: **for** $j \leftarrow 1$ to $|R'|$ **do**
 6: $r_j \leftarrow R'[j]$ // Get current token
 7: $\text{index}_j \leftarrow |\tilde{X}| + j$ // Compute absolute index
 8: **if** r_j is a core token **then** // Only align core tokens
 9: $c_j \leftarrow \text{ApplyMask}(T, \text{index}_j)$
 10: $D_j \leftarrow \text{TopK}\{f_{\text{MLM}}(c_j)\}$ // Fetch top- k candidates
 11: **for** each $w \in D_j$ **do**
 12: $u(w) = \alpha \cdot P_{\text{MLM}}(w | c_j) + (1 - \alpha) \cdot f_{\text{leg}}(w)$
 13: **end for**
 14: $r'_j \leftarrow \arg \max_{w \in D_j} u(w)$ // Select optimal token
 15: $T' \leftarrow \text{Replace}(T, \text{index}_j, r'_j)$ // Update sequence
 16: **end if**
 17: **end for**
 18: $A \leftarrow \text{Suffix}(T', |R'|)$ // Extract the analogy response
 19: **return** A

Reverse mapping: Based on the mapping table π constructed by the Analogy selection module, we perform reverse mapping on the mapped segments within the model-generated response R , obtaining the adjusted response R' .

Semantic concatenation: The original user prompt X is semantically concatenated with the reverse-mapped response R' to form the text T for further processing.

Semantic alignment: To improve the semantic accuracy of the response, we conduct token-level semantic alignment on the preliminary response R' , focusing on core tokens excluding punctuation, determiners, prepositions, and particles. For each token r_j , its context c_j is constructed by masking the j -th position in the full sequence T . The masked language model f_{MLM} then predicts the probability distribution at this position:

$$P_{\text{MLM}}(\cdot | c_j) = f_{\text{MLM}}(T_{[\text{mask}]}). \quad (8)$$

Following the LimitedDomain mechanism [37], the candidate space is restricted to a set D_j consisting of the top- k tokens predicted by the masked language model according to their prediction probabilities. To select a replacement from D_j under the LDP mechanism, we define a utility function $u(w)$ that jointly considers prediction probability and semantic legitimacy in the candidate selection process:

$$u(w) = \alpha \cdot P_{\text{MLM}}(w | c_j) + (1 - \alpha) \cdot f_{\text{leg}}(w), \quad (9)$$

where $\alpha \in [0, 1]$ balances the two terms, and $f_{\text{leg}}(w)$ denotes the semantic validity score of token w . Under the LDP mechanism, a substitute is sampled from D_j according to the utility function $u(w)$. After all protected tokens have been replaced, the response segment is extracted from the updated sequence T to construct the analogy response A , which remains semantically aligned with the original prompt.

C. Response Refinement

We employ a locally deployed lightweight language model to refine the analogy response A , improving its quality and completeness with respect to the user's original query. The refined response is then returned to the user. The prompt used for response refinement is provided in Appendix B, available online.

V. EXPERIMENTS

This section presents a comparative analysis of *PromptAnalogy* against representative text sanitization methods and of the complete *PrivAnalogy* framework against the state-of-the-art LLM privacy framework.

A. Experimental Setup

Implementation: All experiments were implemented in Python. Prediction and substitution of sensitive tokens were performed using a pre-trained BERT model (bert-large-uncased). The experiments were conducted locally on a machine equipped with an Intel i5-12600KF processor and an NVIDIA RTX 4060 Ti GPU. For cloud-based response generation, the GPT-4.1 model was utilized. Response refinement is performed using the locally deployed Mistral-7B model.

Datasets: This study employs the large-scale CNN/Daily Mail dataset [38] and the QNLI benchmark dataset [39] to evaluate various methods on text privacy processing tasks. Supplementary experiments on the SST-2 dataset [40] are included in Appendix D, available online. Furthermore, to verify the practical applicability of the proposed *PrivAnalogy* framework in real-world settings, empirical analyses were conducted on three authentic human-AI dialogue datasets: HC3 [41], Alpaca [42], and UltraChat [43].

Baselines: *PrivAnalogy* is systematically assessed from both privacy and utility aspects. We compare it with representative text sanitization methods, including SANTEXT+ [18], CUSTEXT+ [19], and RANTEXT [12]. To evaluate the end-to-end performance of the overall framework, we integrate these text sanitization methods into InferDPT [12] and use the resulting pipelines as baselines. Following the local inference paradigm of InferDPT, SANTEXT+, CUSTEXT+, and RANTEXT are adapted with a local LLM, enabling a consistent experimental setting for fair comparison.

Evaluation Settings: We evaluate all methods under two settings. The first varies the privacy budget ϵ to analyze the privacy-utility trade-off under different privacy constraints. The second compares methods at controlled levels of semantic preservation. Specifically, we collect protected prompts generated with different privacy budgets and compute the BERTScore [44], [45]

between original and protected prompts as *Semantic fidelity*, group transformed samples according to this score, and report privacy and utility within each interval. This strategy reduces the impact of different perturbation strengths and enables a fairer comparison.

As illustrated in Fig. 4, the distributions of BERTScore for four different methods on the QNLI dataset are presented after text transformation. For a more detailed and equitable comparison, evaluation samples are stratified into four Semantic fidelity intervals: (0.825, 0.85), (0.85, 0.875), (0.875, 0.9), and (0.9, 0.925).

B. Evaluation of Text Protection

In this set of experiments, we comprehensively evaluate the proposed *PromptAnalogy* and compare it with existing data sanitization methods. We first take the privacy budget ϵ as the variable and evaluate the utility-privacy trade-off of different methods on QNLI classification tasks.

We further evaluate privacy and utility across different *Semantic fidelity* intervals on QNLI and CNN/Daily Mail. In each interval, privacy is measured by the defense success rate against the Mask Token Inference Attack [18], while utility is measured by text quality. This setting enables a comparison of the privacy protection capability and data utility of different methods under controlled levels of semantic preservation.

1) *Valuation Under Different Privacy Budgets:* We evaluate all methods under different ϵ values to assess resistance to reconstruction attacks and downstream utility.

Privacy: We evaluate privacy protection on QNLI and SST-2 using four representative reconstruction attacks. The embedding inversion attack [46] and the Inversion BERT attack [47] examine whether sanitized text can be recovered through embedding similarity or neural inversion. We further use the Context-free Bayesian attack and the Contextual Bayesian attack [48] for probabilistic reconstruction. Let ASR denote the attack success rate. We define $1 - \text{ASR}$ as the defense success rate and use it as the privacy metric.

As shown in Fig. 5, on QNLI, *PromptAnalogy* achieves an average defense success rate of approximately 98% across all four attacks, with no noticeable performance degradation as ϵ increases from 1 to 8. In contrast, the performance of CUSTEXT+ declines consistently under the Contextual Bayesian attack, the Context-free Bayesian attack, and the Embedding inversion attack. For example, the defense success rate under the Embedding inversion attack decreases from 0.95 to 0.50. The performance of SANTEXT+ degrades more substantially. Under the Context-free Bayesian attack, the Contextual Bayesian attack, and the Embedding inversion attack, the defense success rate approaches zero as ϵ increases, indicating that the original text can be reconstructed more easily under weaker privacy protection. RANTEXT also maintains a high defense success rate across all attacks, demonstrating that randomized token perturbation effectively weakens the correspondence between sanitized and original text, thereby improving robustness against reconstruction attacks.

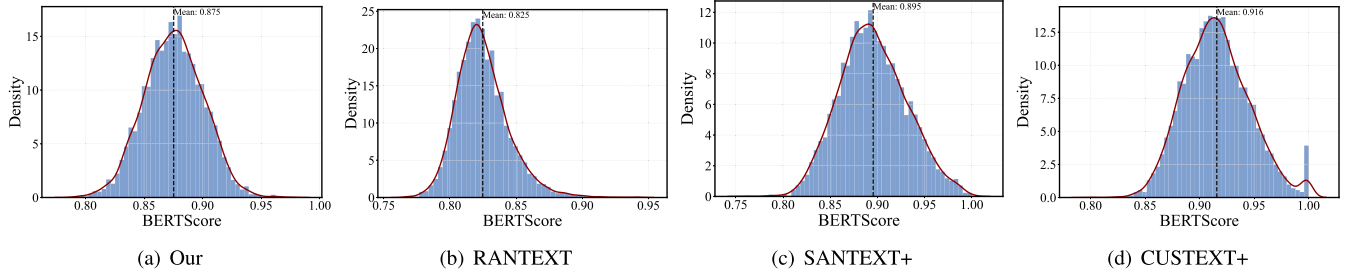


Fig. 4. BERTScore distributions between original and privacy-preserved texts for four evaluation methods on the QNLI dataset.

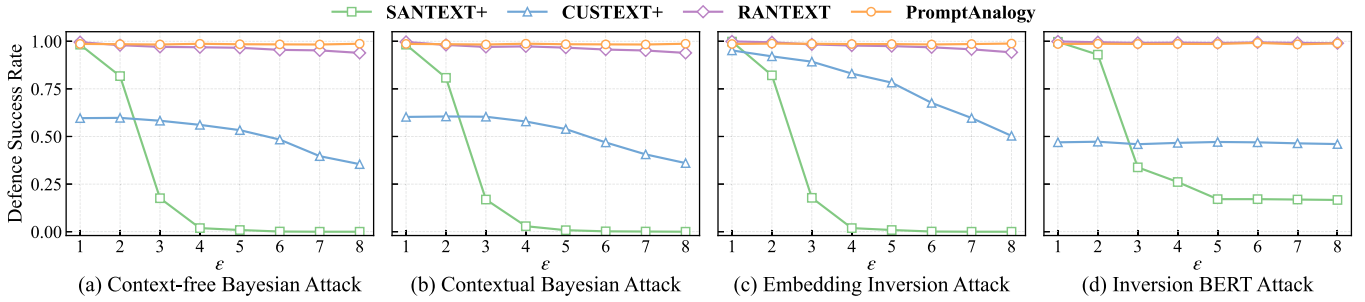


Fig. 5. Privacy performance of four privacy-preserving methods against four reconstruction attacks on the QNLI dataset.

TABLE II
UNCHANGED TOKEN RATIO OF DIFFERENT SANITIZATION METHODS UNDER VARYING PRIVACY BUDGETS

Dataset	Method	Unchanged Token Ratio (%)			
		$\epsilon = 2$	$\epsilon = 4$	$\epsilon = 6$	$\epsilon = 8$
QNLI	SANTEXT+	50.37	98.42	99.86	99.97
	CUSTEXT+	48.53	55.85	65.52	77.79
	RANTEXT	0.87	2.23	4.06	7.30
	PromptAnalogy	51.12	50.68	50.87	50.70

TABLE III
DOWNSTREAM ACCURACY (%) ON QNLI UNDER DIFFERENT PRIVACY BUDGETS

Method	Accuracy $\uparrow$							
	$\epsilon=1$	$\epsilon=2$	$\epsilon=3$	$\epsilon=4$	$\epsilon=5$	$\epsilon=6$	$\epsilon=7$	$\epsilon=8$
SANTEXT+	52.50	63.87	71.86	68.26	66.27	68.46	67.66	69.66
CUSTEXT+	<u>76.45</u>	<u>76.25</u>	<u>77.64</u>	<u>81.04</u>	<u>84.03</u>	87.43	86.43	91.82
RANTEXT	49.30	49.30	49.30	50.50	49.10	49.30	50.10	51.10
PromptAnalogy	84.43	84.23	85.63	86.43	85.43	<u>83.23</u>	<u>86.23</u>	<u>85.00</u>

Similar trends are observed on SST-2, where *PromptAnalogy* consistently demonstrates strong resistance against reconstruction attacks. Since the results on SST-2 follow the same overall trend, we provide the corresponding results in Appendix D, available online to maintain the readability and conciseness of the main text.

Table II provides further insight into the defense performance of the compared methods. SANTEXT+ is highly sensitive to ϵ . When $\epsilon \geq 4$, the unchanged token ratio approaches 100%, making the sanitized text nearly identical to the original text and substantially reducing resistance to reconstruction attacks. CUSTEXT+ also preserves an increasing number of original tokens as ϵ increases, thereby facilitating reverse reconstruction. In contrast, RANTEXT retains almost no original tokens, making direct reconstruction difficult but potentially introducing greater semantic distortion. *PromptAnalogy* maintains a stable unchanged token ratio of approximately 51% on QNLI across different privacy budgets, indicating that its perturbation strength remains largely unaffected by increasing ϵ . As a result, it consistently achieves strong defense performance while preserving semantic fidelity.

Utility: As shown in Table III, on the QNLI downstream task, *PromptAnalogy* consistently achieves an accuracy between

83% and 86%, demonstrating its ability to preserve the semantic relationships required for sentence-pair inference after replacing sensitive semantic associations. The accuracy of CUSTEXT+ increases from approximately 76% to 92% as ϵ increases, indicating that a larger privacy budget preserves more original semantic information and consequently improves downstream task performance. This observation is consistent with the reduced privacy protection demonstrated in the attack experiments. SANTEXT+ achieves an accuracy ranging from approximately 52% to 72%, remaining lower than that of *PromptAnalogy* and CUSTEXT+ under larger privacy budgets. In contrast, the accuracy of RANTEXT remains close to 50%, suggesting that although randomized token perturbation effectively enhances resistance to reconstruction attacks, it substantially degrades the semantic information required for downstream inference.

2) *Evaluation Under Different Semantic Fidelity Levels:* We group privacy-preserved prompts by *Semantic fidelity* and evaluate privacy and utility within each range.

Privacy: We use the mask token inference attack [18] to evaluate privacy. The attack uses a pre-trained BERT model to recover original sensitive tokens from privacy-preserved prompts. Let

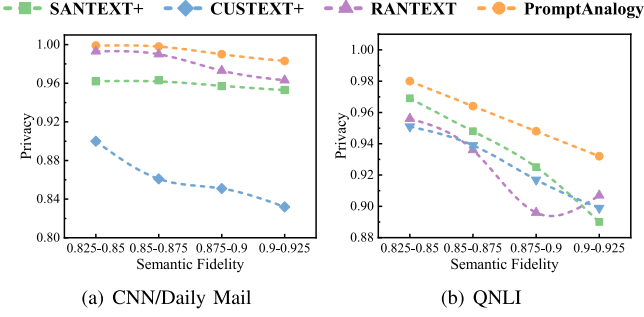


Fig. 6. Defense performance against BERT inference attacks across *Semantic fidelity* intervals for *PromptAnalogy* and three text sanitization methods.

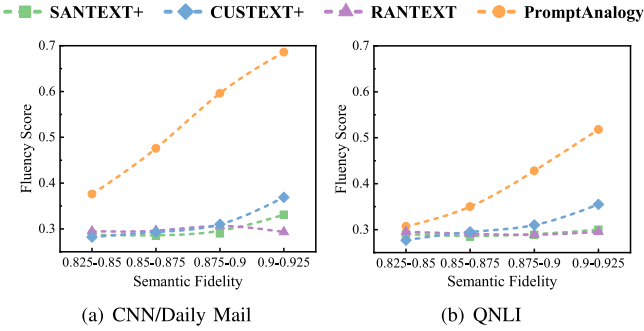


Fig. 7. Fluency Score comparison across *Semantic fidelity* intervals for *PromptAnalogy* and three text sanitization methods.

ASR denote the attack success rate. We use $1 - \text{ASR}$ as the privacy metric.

Fig. 6 reports the results of the BERT inference attack. On CNN/Daily Mail and QNLI, *PromptAnalogy* consistently achieves stronger defense performance than the three text sanitization baselines. Across the four *Semantic fidelity* intervals on both datasets, *PromptAnalogy* blocks about 97% of BERT inference attempts on average. The result also shows that higher BERTScore values tend to correspond to lower privacy scores. This trend suggests that prompts closer to the original text are easier to infer, confirming the trade-off between semantic utility and privacy risk.

Utility: To comprehensively evaluate the linguistic quality of the generated text, we adopt a composite metric that combines language fluency and grammatical correctness. Language fluency is measured using the perplexity computed by the GPT-2 model [49]. Grammatical correctness is evaluated through syntactic analysis with the SpaCy toolkit [50], which detects common grammatical errors, including subject–verb disagreement, incorrect preposition usage, and article misuse. The composite utility score is computed as follows:

$$\text{Fluency Score} = \lambda \cdot f_{\text{Fluency}} + (1 - \lambda) \cdot f_{\text{Grammar}}, \quad (10)$$

where f_{Fluency} denotes the normalized fluency score derived from GPT-2, f_{Grammar} denotes the grammatical correctness score computed by SpaCy, and $\lambda \in [0, 1]$ controls the trade-off between the two components. A higher Fluency Score indicates better overall linguistic quality of the generated text.

As shown in Fig. 7, *PromptAnalogy* outperforms existing mainstream approaches in terms of overall text quality on the CNN/Daily Mail and QNLI datasets. Experimental results indicate that, compared to RANTEXT, the method improves text usability metrics by a factor of 1.6 and achieves the same level of improvement over SANTEXT+, while outperforming CUSTEXT+ by a factor of 1.5. These results suggest that our method generates text with higher readability and expressive accuracy while preserving privacy. Further analysis reveals that as *Semantic fidelity* increases, the Fluency Score also shows a consistent upward trend, indicating that improvements in semantic retention are accompanied by enhanced fluency. This observation supports the validity of *Semantic fidelity* as a metric for assessing semantic utility. The core reason behind these improvements lies in the method’s consideration of both semantic consistency and explicit legitimacy assessment. *PromptAnalogy* jointly considers grammatical structure and contextual coherence, ensuring that substitutions are both semantically appropriate and linguistically natural. This approach significantly enhances the overall quality of the generated text.

C. Overall Framework Evaluation

This set of experiments evaluates the end-to-end performance of *PrivAnalogy* in realistic user–LLM interaction scenarios. Unlike the preceding experiments, which focus on text sanitization and downstream task performance of *PromptAnalogy*, these experiments evaluate the complete prompt protection pipeline, including prompt sanitization, LLM inference, and response restoration. Three real-world human–AI dialogue datasets, Alpaca, HC3, and UltraChat, are used for evaluation. Mistral-7B serves as a local LLM to assess the robustness and generalizability of the framework across different model backends. We compare *PrivAnalogy* with the InferDPT framework and three InferDPT-based variants that employ SANTEXT+, CUSTEXT+, and RANTEXT as the underlying text sanitization methods.

This study evaluates privacy performance on three real-world user–AI dialogue datasets: Alpaca, HC3, and UltraChat. The experiments are conducted across different *Semantic fidelity* intervals to analyze the relationship between privacy protection and semantic preservation. For CUSTEXT+, no samples fall within the $(0.825, 0.85)$ *Semantic fidelity* interval on the Alpaca and UltraChat datasets. Therefore, no results are reported for this method in that interval. Additional experiments on different privacy budgets and the generalizability evaluation across different local models are provided in Appendix E, available online.

Privacy: To evaluate the risk of prompt leakage during interactions with cloud-based LLMs, we introduce a Prompt inversion attack as a privacy assessment method. Specifically, GPT-4.1 is used to infer the original prompt from the response generated by the cloud-based LLM. The prompt template used for the Prompt inversion attack is provided in Appendix C, available online. To quantify privacy leakage, we compute the cosine similarity between the inferred prompt X' and the original prompt X :

$$\text{Sim}(X, X') = \frac{E(X) \cdot E(X')}{\|E(X)\| \|E(X')\|}, \quad (11)$$

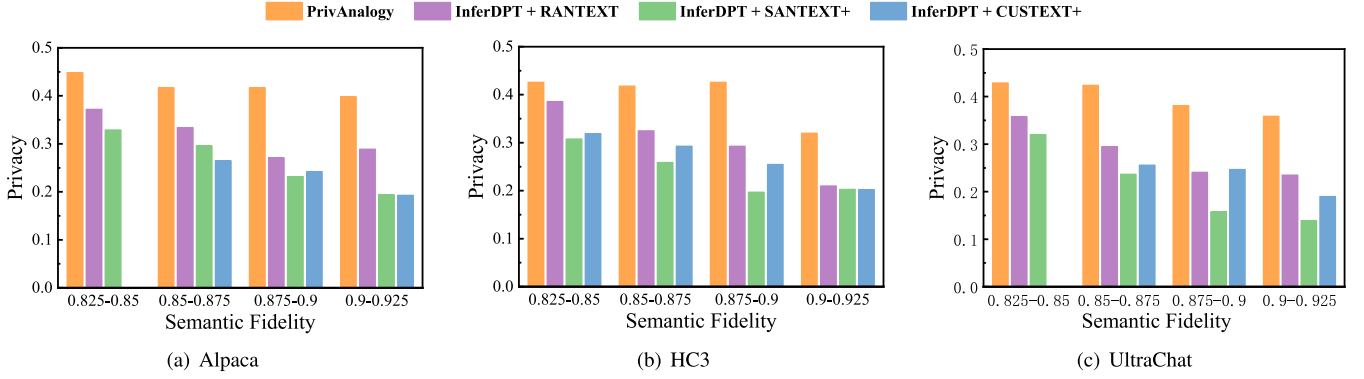


Fig. 8. The test results of defending against Prompt inversion attacks using *PrivAnalogy* and three baseline methods based on Alpaca, HC3, and UltraChat datasets.

where $E(\cdot)$ denotes the text embedding function. The privacy score is then defined as

$$\text{Privacy} = 1 - \frac{\text{Sim}(X, X') + 1}{2}, \quad (12)$$

where a higher privacy score indicates a lower degree of prompt leakage.

As shown in Fig. 8, *PrivAnalogy* provides stronger privacy protection against Prompt inversion attacks compared to other methods. This result further validates the privacy from the perspective of model-generated responses. Previously, our BERT inference attack experiments also demonstrated that texts processed by *PromptAnalogy* exhibit enhanced privacy levels. Since those experiments are conducted based on all information accessible to the cloud-based black-box LLM, they offer strong empirical support.

This observation is also consistent with the additional privacy-budget experiments reported in Appendix E, available online (Table VIII). *InferDPT + RANTEXT* usually obtains higher privacy scores across privacy budgets because its stronger random perturbation weakens the semantic correspondence between the original and protected prompts. Here, samples are grouped by *Semantic fidelity*, which evaluates privacy under similar semantic-preservation levels and reduces the effect of perturbation strength. Under this controlled setting, *RANTEXT* lacks explicit modeling of sensitive associations and task-specific semantic structure, so high-fidelity samples still retain cues related to the original intent. In contrast, *PromptAnalogy* changes the semantic mapping of sensitive information through sensitive association identification and analogy selection, weakening the reversible relation between cloud-visible content and the original sensitive semantics. Therefore, within the same *Semantic fidelity* interval, *PrivAnalogy* provides stronger defense against Prompt inversion attacks in realistic deployment scenarios.

Utility: This subsection evaluates utility from two complementary perspectives: response accuracy and linguistic coherence. Accuracy quantifies the extent to which the rewritten response preserves the information required to answer the original user prompt, whereas coherence quantifies the linguistic

consistency of the resulting prompt–response interaction. Together, these metrics evaluate whether the proposed framework preserves response quality while protecting prompt privacy.

a) Accuracy. Accuracy is measured using the BERTScore F1 score between the reference response y_i and the final response $\hat{y}_i$:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \text{BERTScore}_{F1}(y_i, \hat{y}_i). \quad (13)$$

b) Coherence. Coherence is measured using the perplexity of the concatenated prompt–response sequence $z_i = [x_i; \hat{y}_i]$, where x_i denotes the original user prompt and $\hat{y}_i$ denotes the rewritten response. To obtain a bounded metric, the perplexity is clipped to the range [10,1000] and linearly normalized. The overall coherence is computed as:

$$\text{Coherence} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{\text{clip}(\text{PPL}(z_i), 10, 1000) - 10}{990} \right). \quad (14)$$

Table IV reports Accuracy and Coherence when Mistral-7B is used as the local LLM. The results show that *PrivAnalogy* can generate useful responses after analogy reversion and response refinement. Together with the privacy results reported above, these utility results suggest that *PrivAnalogy* strengthens prompt protection while preserving response quality, yielding a practical trade-off between protection and utility.

Ablation Study: Table V compares the complete *PrivAnalogy* framework with *PrivAnalogy* w/o AR, a variant without the analogy reversion module, using Mistral-7B as the local LLM. Across different *Semantic fidelity* settings, *PrivAnalogy* generally achieves better *Accuracy* and *Coherence* than *PrivAnalogy* w/o AR across all three datasets. These results demonstrate that the analogy reversion module effectively recovers task-relevant semantics affected by prompt transformation, thereby improving response correctness and semantic consistency. In contrast, *PrivAnalogy* w/o AR directly uses responses generated from transformed prompts, making it more vulnerable to semantic deviations introduced by analogy transformation.

TABLE IV
COMPARISON OF ACCURACY AND FLUENCY ACROSS DIFFERENT *SEMANTIC FIDELITY* INTERVALS ON ALPACA, HC3, AND ULTRACHAT, USING MISTRAL-7B AS THE LOCAL LLM. DASHES (–) INDICATE UNAVAILABLE RESULTS

Dataset	Method	Accuracy $\uparrow$				Coherence $\uparrow$			
		0.825–0.85	0.85–0.875	0.875–0.9	0.9–0.925	0.825–0.85	0.85–0.875	0.875–0.9	0.9–0.925
Alpaca	InferDPT + SANTEXT+	0.856	0.851	<u>0.857</u>	0.852	<u>0.970</u>	0.957	0.955	<u>0.971</u>
	InferDPT + CUSTEXT+	–	0.851	0.856	<u>0.856</u>	–	0.961	<u>0.976</u>	0.961
	InferDPT + RANTEXT	<u>0.857</u>	<u>0.853</u>	0.848	0.853	0.955	<u>0.968</u>	0.974	0.956
	PrivAnalogy	0.865	0.862	0.861	0.860	0.974	0.979	0.977	0.977
HC3	InferDPT + SANTEXT+	0.840	<u>0.856</u>	0.856	0.847	0.970	<u>0.974</u>	<u>0.973</u>	0.972
	InferDPT + CUSTEXT+	0.833	0.846	0.853	0.873	0.967	0.969	0.969	<u>0.973</u>
	InferDPT + RANTEXT	0.861	0.853	<u>0.859</u>	0.858	<u>0.976</u>	0.972	<u>0.973</u>	<u>0.973</u>
	PrivAnalogy	<u>0.841</u>	0.864	0.868	<u>0.864</u>	0.978	0.981	0.981	0.978
UltraChat	InferDPT + SANTEXT+	0.849	<u>0.868</u>	0.871	0.871	<u>0.980</u>	0.978	0.978	0.979
	InferDPT + CUSTEXT+	–	0.867	0.868	<u>0.874</u>	–	0.977	0.975	0.976
	InferDPT + RANTEXT	<u>0.863</u>	0.871	0.873	0.868	0.978	0.981	0.978	0.974
	PrivAnalogy	0.875	0.865	<u>0.872</u>	0.876	0.986	0.982	0.980	<u>0.977</u>

TABLE V
ABLATION STUDY OF THE ANALOGY REVERSION MODULE UNDER DIFFERENT *SEMANTIC FIDELITY* WITH MISTRAL AS THE LOCAL LLM

Dataset	Method	Accuracy $\uparrow$				Coherence $\uparrow$			
		0.825–0.85	0.85–0.875	0.875–0.9	0.9–0.925	0.825–0.85	0.85–0.875	0.875–0.9	0.9–0.925
Alpaca	PrivAnalogy w/o AR	0.864	0.860	0.857	0.855	0.908	0.969	0.970	0.961
	PrivAnalogy	0.865	0.862	0.861	0.860	0.974	0.979	0.977	0.977
HC3	PrivAnalogy w/o AR	0.830	0.856	0.861	0.853	0.969	0.972	0.975	0.975
	PrivAnalogy	0.841	0.864	0.868	0.864	0.978	0.981	0.981	0.978
UltraChat	PrivAnalogy w/o AR	0.872	0.866	0.868	0.875	0.976	0.978	0.980	0.980
	PrivAnalogy	0.875	0.865	0.872	0.876	0.986	0.982	0.980	0.977

TABLE VI
AVERAGE LOCAL RUNTIME OVERHEAD ON ALPACA IN SECONDS. RESPONSE PROCESSING EXCLUDES LOCAL LLM EXTRACTION

Method	Dictionary	Prompt	Response
InferDPT + SANTEXT+	0.0167	0.1555	–
InferDPT + CUSTEXT+	1192.7126	0.0003	–
InferDPT + RANTEXT	8864.2719	0.5076	–
PrivAnalogy	–	0.8138	3.5651

D. Runtime Overhead

Table VI reports the average local runtime overhead in end-to-end interaction scenarios. The runtime characteristics differ substantially across the compared methods. Among the InferDPT-based methods, the overhead of dictionary construction ranges from negligible for InferDPT + SANTEXT+ to dominant for InferDPT + CUSTEXT+ and InferDPT + RANTEXT. In contrast, *PrivAnalogy* does not require offline dictionary construction. Instead, it performs response restoration after LLM inference, making response processing the primary source of local runtime overhead.

VI. LIMITATIONS AND FUTURE WORK

Although *PrivAnalogy* provides a strong privacy–utility trade-off across datasets, its utility may drop when analogy selection alters task-relevant semantic cues. This problem is especially visible in sentiment analysis tasks such as SST-2, where

predictions often hinge on a few polarity-bearing words or phrases. When such positions are treated as sensitive semantic associations and replaced with analogical expressions, the transformed text may dilute or even shift the original sentiment signal. This explains why *PrivAnalogy* still offers strong privacy protection but does not obtain the highest Accuracy on SST-2.

Second, *PrivAnalogy* relies on a local masked language model to generate analogy candidates for protected positions. Although masked language models can provide context-aware candidates in general scenarios, their generation quality may degrade when handling specialized terminology or knowledge-intensive prompts. This limitation is more evident in HC3, which contains domain-specific questions from areas such as medicine and finance. Our analysis of intermediate outputs reveals that some samples contain semantically mismatched analogy candidates or candidates that deviate from the original domain context. These cases may introduce semantic distortion and reduce the utility of the generated responses.

Third, the local masked language model introduces additional computational cost. *PrivAnalogy* dynamically generates context-dependent candidates with a masked language model, filters invalid candidates, and performs LDP-based selection. It also applies analogy reversion after receiving the external model response. As a result, *PrivAnalogy* may incur higher local computation cost and response latency, especially for long prompts, prompts with multiple sensitive semantic associations, or interactions requiring frequent candidate updates.

These limitations suggest several directions for future research. First, incorporating task-aware semantic constraints into analogy selection could better preserve task-relevant information, particularly for classification tasks that rely on a small number of critical semantic cues. Second, adapting the local masked language model to domain-specific corpora could improve the quality of generated analogy candidates for specialized terminology and knowledge-intensive prompts. Finally, more efficient candidate generation and response restoration strategies could reduce local runtime overhead while maintaining the privacy guarantees of *PrivAnalogy*.

VII. CONCLUSION

This paper presents *PrivAnalogy*, a privacy-preserving framework for user prompts based on the Analogy mechanism. The proposed framework protects sensitive semantic associations while preserving the inference capabilities of remote LLMs. Specifically, *PromptAnalogy* integrates the principles of differential privacy with the Analogy mechanism and employs the Gumbel-Max mechanism to enable privacy-preserving text transformation. Compared with conventional text sanitization methods, *PromptAnalogy* achieves stronger privacy protection while maintaining high text utility. Furthermore, an Analogy reversion method is introduced to restore the semantics of model responses and preserve consistency with the original user prompts. Future work will further investigate more effective and efficient privacy-preserving mechanisms for LLM interactions. It is hoped that this work provides useful insights into LLM security and serves as a reference for future research in this area.

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